

Alvaro Perdomo Gutierrez

DevOps Engineer III

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PROFESSIONAL SUMMARY

DevOps Engineer with **12+** years across healthcare, fintech, and gaming infrastructure, specializing in Kubernetes, Docker, and CI/CD automation. Designed and deployed Infrastructure as Code solutions, automated monitoring systems, and continuous delivery pipelines serving millions of users. Expert in cloud platforms, alerting systems, and operational optimization to reduce toil and enable engineering productivity.

SKILLS

Languages: Go, Python, Bash, JavaScript, TypeScript, SQL, HTML, CSS
Infrastructure & IaC: Kubernetes, Docker, Terraform, Helm, CloudFormation, Docker Compose, Infrastructure as Code
CI/CD & Deployment: Jenkins, GitHub Actions, GitLab CI, Bamboo CI, ArgoCD, Spinnaker, Continuous Integration, Continuous Deployment
Monitoring & Observability: Prometheus, Grafana, Datadog, New Relic, ELK Stack, Elasticsearch, Splunk, CloudWatch
Cloud Platforms: AWS, Google Cloud Platform, Azure, EC2, S3, RDS, Lambda, CloudFront, VPC
Database: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, TcaplusDB
Testing: Jest, Pytest, Selenium, Cypress, k6, JMeter
Version Control & Tools: Git, GitHub, GitLab, Jira, Confluence
Others: RESTful API, GraphQL, WebSocket, Linux, Nginx, Apache, SSL/TLS, Agile, Scrum

EXPERIENCE

Leanware May 2024–Present
Senior DevOps Engineer

- Led the design and implementation of Infrastructure as Code using **Terraform** and **Helm**, provisioning multi-tenant environments across **AWS** and **Google Cloud Platform** to support six industry verticals including healthcare, wellness, and financial services platforms.
- Spearheaded the migration of legacy monolithic deployments to containerized microservices on **Kubernetes**, reducing deployment time from **3 hours to 18 minutes** and enabling developers to push changes 10 times daily across staging and production.
- Championed a comprehensive monitoring and alerting strategy using **Prometheus**, **Grafana**, and **Datadog**, instrumenting all 12 platform services and eliminating alert fatigue by implementing intelligent threshold-based notifications that reduced false positives by **87%**.

- Automated the entire continuous integration and deployment pipeline with **Jenkins** and **GitHub Actions**, implementing infrastructure-as-code pattern validation and automated security scanning that caught **23 infrastructure misconfigurations** before reaching production.
- Collaborated with application teams to establish SLOs and SLAs for critical services, designing self-healing mechanisms in **Kubernetes** that automated failover and recovery, cutting mean time to recovery from **45 minutes to 6 minutes** during regional outages.
- Engineered a disaster recovery framework using **AWS** multi-region failover and automated backup validation, achieving **99.95% uptime** across all healthcare-related services while maintaining compliance with HIPAA audit requirements.
- Optimized container image sizes and **Kubernetes** resource requests by profiling production workloads, reducing infrastructure costs by **32%** and enabling the platform to handle 40% more concurrent users without additional hardware provisioning.
- Established centralized logging with the **ELK Stack**, collecting logs from 200+ services and making historical debugging 15x faster for oncall engineers investigating production incidents.
- Mentored four junior engineers on DevOps best practices, containerization strategies, and Infrastructure as Code patterns, resulting in all four shipping production changes independently within 8 weeks.
- Conducted postmortem analysis on two major production incidents, identifying root causes in deployment orchestration and implementing preventive controls that eliminated similar failures.
- Automated environment provisioning for QA and staging, reducing time to spin up a complete test environment from 2 days to under 15 minutes, enabling 5x faster test cycle iterations.
- Built self-service dashboards in **Grafana** that gave product and backend teams real-time visibility into system health metrics, reducing their dependency on DevOps for operational status checks by **68%**.

ThredUp

Mar 2022–May 2024

Senior Software Engineer, Platform Infrastructure

- Drove the redesign of the data platform infrastructure, migrating from on-premises Spark clusters to **Kubernetes**-orchestrated **Docker** containers on **AWS**, which reduced model training time from 8 hours to 2.5 hours and enabled weekly instead of monthly ML model deployments.
- Implemented Infrastructure as Code using **Terraform** for the entire **AWS** estate, managing 150+ resources and reducing manual configuration drift incidents from 8 per month to zero while enabling reproducible production deployments.
- Architected a **Jenkins**-based continuous integration system with **Docker** image caching and parallel test execution, accelerating build times from 28 minutes to 6 minutes and unblocking developers blocked on slow CI feedback.
- Designed and deployed comprehensive monitoring using **New Relic** and custom **Prometheus** metrics, adding observability to the ML pipeline that surfaced a memory leak causing 15% of model training failures, which the team then resolved.
- Collaborated closely with machine learning engineers to establish reproducible environments using **Docker** and version-controlled dependencies, reducing "works on my machine" issues and accelerating onboarding of new ML platform users by 3 weeks.

- Automated the deployment of containerized services across dev, staging, and production using **ArgoCD**, implementing GitOps workflows that gave developers visibility into every deployment and reduced rollback times from 1 hour to 5 minutes.
- Established on-call runbooks and alerting thresholds for 8 critical platform services, reducing mean time to resolution for production incidents from 90 minutes to 18 minutes and improving platform SLA attainment to **99.8%** from **94%**.
- Optimized **Kubernetes** networking and cluster resource utilization, reducing monthly **AWS** infrastructure costs by **28%** while maintaining low-latency data pipeline performance serving 100,000+ daily transactions.
- Implemented automated security scanning in the CI pipeline using **Docker** image analysis tools, blocking 12 known vulnerabilities from reaching production and achieving zero critical CVEs in deployed containers.
- Built self-healing automation in **Kubernetes** that automatically restarted failed pods and rescheduled workloads during node failures, reducing manual intervention during operational incidents by **95%**.

Spring Health

Nov 2020–Mar 2022

Software Engineer, Infrastructure & Platform

- Built scalable infrastructure on **AWS** using **Terraform** and **CloudFormation** to support 6x growth in user load while maintaining sub-100ms API response times and preserving HIPAA compliance across all Fortune 500 customer deployments.
- Implemented comprehensive logging and monitoring for the care navigation platform using **Datadog** and custom **Prometheus** exporters, enabling the operations team to diagnose latency issues across the global infrastructure and fix a **2-second delay** in critical patient matching queries.
- Automated deployment pipelines with **Jenkins** and **Docker**, reducing the time to deploy new versions from 45 minutes to 8 minutes and enabling the platform to roll out weekly security patches without manual intervention.
- Designed and deployed multi-region failover infrastructure on **AWS** using health checks and automated traffic rerouting, ensuring continuity during a datacenter-level incident that would have caused 3+ hours of downtime under legacy systems.
- Collaborated with the security team to implement **SSL/TLS** certificate management and key rotation automation, eliminating the risk of expired certificates and reducing the annual security audit effort by 40 hours.
- Established SLOs and monitoring for four critical backend services, implementing alert escalation policies that woke on-call engineers only for true issues, reducing false alert pages from 12 per week to 2 per week.
- Mentored two junior backend engineers on operational best practices, infrastructure testing patterns, and postmortem analysis, enabling them to diagnose and resolve infrastructure-related bugs independently.
- Optimized container resource requests in **Kubernetes** by profiling production workloads, reducing infrastructure costs by **24%** and enabling the platform to scale to support 2M+ members without proportional cost increases.

Tencent Games

Nov 2014–Nov 2020

Backend Infrastructure Engineer

- Architected distributed backend services for Honor of Kings using **Docker** and containerization principles, supporting 100M+ daily active users with single-digit millisecond response times by implementing intelligent load balancing across geographically distributed data centers.
- Designed and deployed a comprehensive monitoring system using custom **Prometheus** collectors and **Grafana** dashboards, instrumenting all game servers and reducing incident detection time from 15 minutes to 90 seconds, enabling faster response to service degradations.
- Implemented automated deployment pipelines for game server updates using Jenkins and custom orchestration tools, enabling developers to push 50+ builds per day while maintaining **99.9% uptime** during peak gameplay.
- Optimized database infrastructure for TcaplusDB (handling 40M+ daily requests), implementing intelligent caching strategies with **Redis** and query optimization that reduced average response time from 150ms to **35ms**, improving player experience measurably.
- Deployed anti-cheat kernel-level monitoring at scale across 8M+ concurrent game servers, integrating with **Docker** container security features to minimize false positives while blocking 99.7% of known cheat detection vectors.
- Collaborated with platform engineers to establish operational runbooks and incident response procedures for 12 live game services, reducing mean time to resolution for critical issues from 2 hours to 12 minutes through automation and clear escalation paths.
- Built centralized logging infrastructure using **ELK Stack** for game telemetry, processing 500M+ events daily and enabling analytics teams to debug player behavior issues and optimize game balance.

EDUCATION

The Hong Kong Polytechnic University

Sep 2010–Nov 2014

Bachelor of Science in Computer Science